



Launching VPC Resources

IR

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Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation [Info](#)
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate
nextwork

IPv4 CIDR block [Info](#)
Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16 65,536 IPs

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block

Tenancy [Info](#)
Default

► Encryption settings - optional

Preview

Subnets (6)
Subnets within this VPC

us-east-2a

- nextwork-subnet-public1-us-east-
- nextwork-subnet-private1-us-east-
- nextwork-subnet-private3-us-east-

us-east-2b

- nextwork-subnet-public2-us-east-
- nextwork-subnet-private2-us-east-
- nextwork-subnet-private4-us-east-

Route tables (5)
Route network traffic to resources

- nextwork-rtb-public
- nextwork-rtb-private1-us-east-2a
- nextwork-rtb-private2-us-east-2b
- nextwork-rtb-private3-us-east-2a
- nextwork-rtb-private4-us-east-2b



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is a service that lets you create a private, isolated network in AWS where you can launch and manage resources like EC2 instances. It is useful because it gives you control over networking, including IP addresses, subnets, route tables, and security, allowing you to securely organize and isolate your cloud infrastructure.

How I used Amazon VPC in this project

I used Amazon VPC to create and configure public and private subnets, route tables, internet gateways and security groups.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was the absolute simplicity and straightforward guidance.

This project took me...

This project took me around 30 minutes to complete



Setting Up Direct VM Access

Directly accessing an EC2 instance means logging into the virtual server itself (using SSH for Linux or RDP for Windows) so you can manage its operating system, files, and software.

SSH is a key method for directly accessing a VM

SSH traffic means "Secure Shell" and it is a way that developers can use to directly log in to a remote computer from their machines.

To enable direct access, I set up key pairs

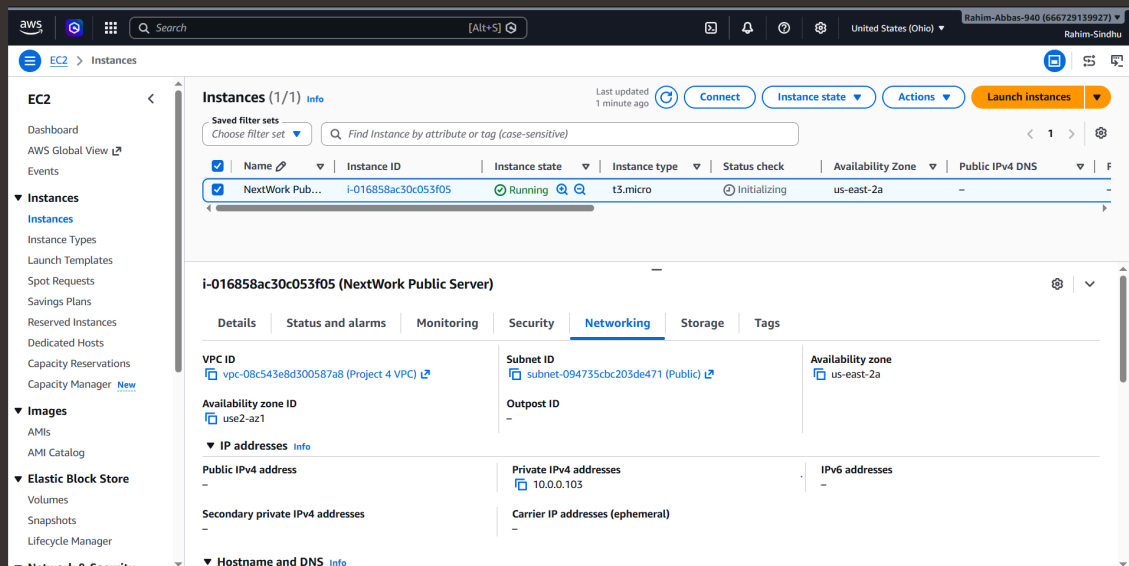
Key pairs are a pair of public and private keys that provide secure access to your EC2 instance. The public key is stored on the EC2 instance, while the private key is kept securely by the user. When the user attempts to connect to the instance, they prove possession of the private key through a cryptographic authentication process. The instance verifies this using the stored public key, and access is granted only if the keys match.

A private key's file format means "Privacy-Enhanced Mail" which is a text-based file format used to store cryptographic keys and certificates, such as an EC2 private key. My private key's file format was ".pem".



Launching a public server

I had to change my EC2 instance's networking settings by modifying its associated security groups, subnet, route table, and network ACL settings within the VPC to allow the required inbound and outbound traffic.





Launching a private server

My private server has its own dedicated security group because the private server needs stricter access control, it is not exposed to the internet, while the public server allows internet traffic, so they use different security groups to apply different security rules.

My private server's security group's source is the security group of my public server so that only traffic coming from the public server is allowed, keeping the private server inaccessible from the internet while still enabling internal communication between them.

The screenshot displays the AWS Management Console interface for launching an EC2 instance. The 'Firewall (security groups)' step is selected, showing options to 'Create security group' or 'Select existing security group'. The 'Create security group' option is chosen, and the 'Security group name' is set to 'NextWork Private Security Group'. The 'Description' is 'Security group for NextWork Private Subnet'. Under 'Inbound Security Group Rules', a rule is added with 'Type' set to 'ssh', 'Protocol' set to 'TCP', 'Port range' set to '22', 'Source type' set to 'Custom', and 'Source' set to 'sg-0bdcfadd36a8e8df5'. The 'Add security group rule' button is visible. The 'Summary' panel on the right shows the configuration for 1 instance, using the 'Amazon Linux 2023 AMI' (ami-0x6ac5f2fed2981b0), 't3.micro' instance type, and 8 GiB storage. The 'Launch instance' button is highlighted in orange.

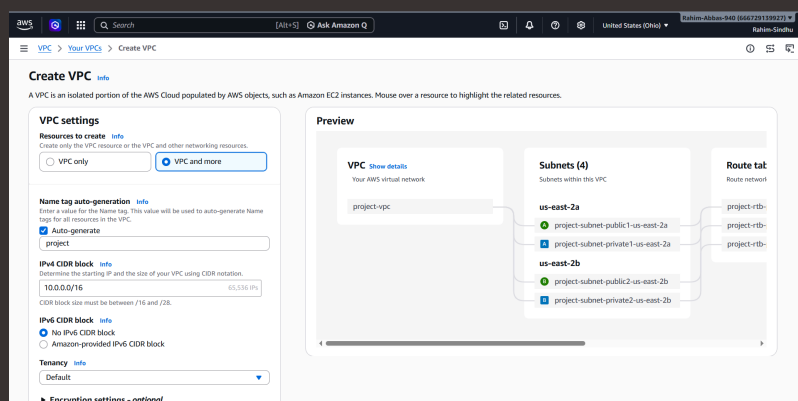


Speeding up VPC creation

I used an alternative way to set up an Amazon VPC! This time, I used a resource map to help me in mapping my VPC with public and private subnets, route tables and the internet gateways.

A VPC resource map is a visual diagram that shows all your AWS resources within a VPC and how they are connected to each other.

My new VPC has a CIDR block of 10.0.0.0/16. It is possible for it to have the same IPv4 CIDR block as an existing VPC because VPCs are logically isolated from each other in AWS, so overlapping CIDR ranges do not conflict unless the VPCs are connected through peering or other networking setups.



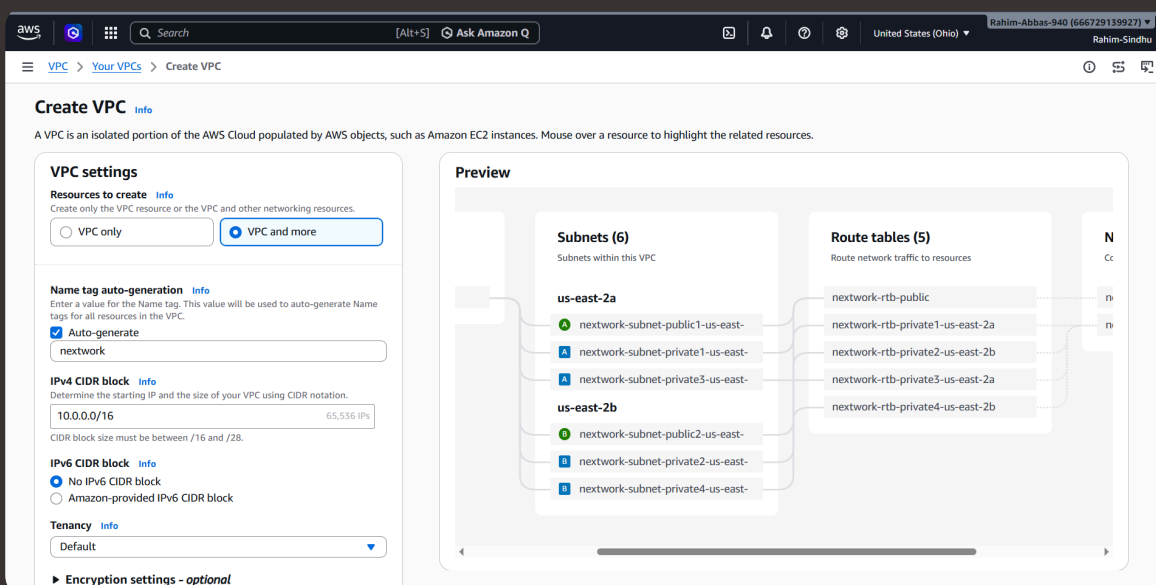


Speeding up VPC creation

Tips for using the VPC resource map

When determining the number of public subnets in my VPC, I only had two options: 0 or 2. This was because I created a subnet in each Availability Zone for high availability, and each public subnet is tied to a specific AZ, so the template or lab required either not creating any public subnets or creating one per AZ (two AZs = two public subnets).

A NAT Gateway is an AWS service that allows instances in a private subnet to access the internet for outbound traffic, while still preventing the internet from initiating connections to them. It is used so private servers can: - Download updates - Access external APIs - Install packages but remain not publicly accessible.





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